

T≡M Theory — Time Is Motion: Cosmological Expansion as the Root Tick

Time as Hierarchical Motion Nested within Cosmological Expansion (Root Tick ☺)

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Dedicated to: my son and my wife, my guides in the Tempo.

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Author's note

I do not come from the scientific or academic world. This theory comes from a long-held intuition: **time is motion**. In my v1.0 I tried to express it with words. In v2.0 I tried to express it with a more technical structure without losing my way of speaking.

v2.1 came about because v2.0 did what it was supposed to do: it invited review, and the review found something real. The minimal ansatz of v2.0, in its simplest form, turned out to have **no observable content** — the residual parameter cancels in the only quantity metrology actually measures. That version corrected this, adopted the minimal form that *does* have observable content, and translated existing experimental constraints into the **first numerical bound** on the T≡M residual parameter.

This v2.2 adds the link that was missing between the premise and the experimental program: the closure condition. It is what gives “time is motion” a consequence, rather than leaving it a stated identity.

As in previous versions, the order, notation, and formalization were produced with AI assistance, because my expertise in physics is limited. The idea, the conviction, and the decisions about what to keep are

mine; the tool helps me say them with more rigor. I do not present this as a closed result, but as the next step of an open process: corrections, counterexamples, and proposals remain welcome.

What changed in v2.2 (changelog)

1. **New postulate: $T \equiv M-4$ (closure).** Not every motion constitutes proper time: the motion must close on itself. This is what gives the identity its content, and in earlier versions it was only implicit.
2. **New section: “Closure and proper time.”** Develops the chain that runs from the premise to the experimental program, with the case of the photon as its first consequence.
3. **New section: “What it would take to abandon $T \equiv M$.”** An explicit refutation criterion.
4. **Expanded caveats.** It is now stated that the known local effect of expansion on bound systems is a static offset and not a cumulative drift, also under quantum treatment (Price and Romano, 2012; Carrera and Giulini, 2010; Lemos, 2014), and that the two-factor form of eq. (9) coincides with the standard structure of the phenomenology of coupled light scalars (Trivedi, 2025).
5. **Minimal program:** a point is added on pairs of transitions within a single species.

Everything else — $T \equiv M-0$ to $T \equiv M-3$, the definition of Θ , the nested structure, the minimal event, the strong conjecture, eq. (8), bound (11), and the honest limit — is unchanged from v2.1.

Reminder of what v2.1 introduced, for readers arriving directly at this version: it showed that the *constant*-residual ansatz of v2.0 was unobservable — a universal residual cancels in frequency ratios, and a clock-dependent constant one is absorbed into calibration — so the weak branch moved to a **drift** form with accumulated expansion, and the first numerical bound on the differential residual was obtained.

Abstract

I propose a framework where **Time (Tempo)** is not an independent entity, but is **identical to universal motion**. The **root tick** of Tempo is cosmological expansion, represented as accumulated expansion Θ . All local motions exist as **nested motions** inside that root tick: different clocks measure different “elapsed time” because they do not measure an external container, but **internal motion counts**.

This v2.2 makes explicit the condition that gives that identity its content: **not every motion constitutes proper time, only the one that closes on itself**. It follows that systems with different closures might couple differently to the root tick, which is the conceptual origin of the residual parameter.

In v2.1 the weak (testable) branch was corrected and sharpened. It was shown that the constant-residual ansatz of v2.0 is unobservable in frequency-ratio measurements; the minimal observable extension is a residual **drift** of clock rates with accumulated expansion, parametrized by ε_R per unit Θ . Translating existing long-term optical-clock ratio constraints (fractional drifts $\lesssim 10^{-18} \text{ yr}^{-1}$) through today’s expansion rate ($H_0 \approx 7 \times 10^{-11} \text{ yr}^{-1}$) yields the first numerical bound on the differential residual:

$$|\Delta\varepsilon_R| \lesssim 10^{-8}$$

A universal residual remains unobservable in ratios and is explicitly flagged as such. $T \equiv M$ thus stands as an ontological/synthesis framework whose weak experimental branch is now quantitatively bounded — which is itself a result.

What is conviction and what is experiment

This manuscript mixes three layers, and I separate them so we don't confuse them:

- **Conviction / Ontology:** how I interpret what time is (my base idea: Tempo $\equiv$ motion).
 - **Synthesis:** a conceptual re-organization using language compatible with known physics (using Θ as the root tick, clocks as counts).
 - **Experiment (weak hypothesis):** a minimal extension that, if it exists, can be bounded by data — in its corrected, observable form (eq. 8, v2.1), with a first bound (eq. 11).
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Central idea (no detours)

Tempo $\equiv$ universal motion.

And the **root motion** (the most fundamental tick) is **cosmological expansion**.

Everything else is **nested motion** inside that root: subfolders inside a parent folder.

T $\equiv$ M postulates (the minimum I claim)

T $\equiv$ M-0 — Identity

Time (Tempo) is identical to the motion/change of the universe; it does not exist as a separate entity.

T $\equiv$ M-1 — Root tick

The **root tick** of Tempo is identified with cosmic becoming: the **expansion of the universe**.

T $\equiv$ M-2 — Nested motion

Local motions are **nested motions**: they occur inside the advance of the root tick, across levels (cosmological, astrophysical, planetary, biological, microscopic, etc.).

T $\equiv$ M-3 — Minimal event and no-pause

There exists a form of **minimal activity** required for local change to exist. If that minimal activity is null, local change is null (**total pause**).

T $\equiv$ M-4 — Closure

Not every motion constitutes proper time. For a motion to generate a proper time, it must **close on itself**: its internal relations must sustain one another, forming a unit. **Motion is the dynamical condition; closure is the condition for that motion to constitute a proper time.**

A motion that does not close exists and propagates, but accumulates no count of its own.

Closure and proper time

T $\equiv$ M-0 says that time is motion. T $\equiv$ M-4 adds the condition that gives it content: not just any motion, but the one that closes. This section develops what follows from that.

The first consequence: what does not close has no proper time

Light does not close. It propagates at the limiting speed, and at that limit nothing can catch up with it — not even what would have to return on itself to complete an internal circuit. Under T $\equiv$ M-4, then, a coherence that only propagates accumulates no count of its own.

Special relativity says the same thing by an entirely different route: along a null trajectory, proper time is zero.

On the wording: I use “zero proper time along a null trajectory,” which is the correct formulation. There is no reference frame comoving with light, so expressions such as “the photon does not experience time” are imprecise, even if they point to the same thing.

That the closure condition reproduces this established fact without having been designed to is the cleanest consequence of $T \equiv M$. And it is what distinguishes a thesis from a definition: a definition predicts nothing; this one predicts that without closure there is no proper time, and that turns out to be true.

There is a second place where the condition has a known correlate. Most of the mass of ordinary matter does not come from the masses of its constituents but from confined field energy, much of it associated with massless fields. Under $T \equiv M-4$ this admits a direct reading — what remains confined closes, and has proper time — although *the reading belongs to this framework and is not a result of particle physics*, and it does not extend to elementary particles, whose mass comes from a different mechanism. I note it as a correspondence, not as an argument.

The chain

With $T \equiv M-4$ in place, what runs from the premise to the experimental program of this manuscript can be stated in one piece:

Time is motion. But not every motion constitutes proper time: it must close. **The photon does not close, and has no proper time.** If proper time depends on closure, then systems with different types of closure might couple differently to the root tick. **That difference in coupling is what the residual parameter ε_R tries to capture**, and it is what one looks for by comparing clocks.

The rest of this manuscript develops the last link. The first three are the reason the last one makes sense.

It is worth marking where each thing stands. $T \equiv M-4$ is a postulate. The case of the photon is its consequence, and it holds. That there exist types of closure with different couplings is a **hypothesis**, and this manuscript does not develop it: what exactly distinguishes two closures of different type is an open question that I flag as such.

A further caution about the term. “Closure” still covers several notions that in physics would be distinct — a closed trajectory, a dynamically bound system, a self-consistent configuration — and this manuscript does not distinguish them. It is a **conceptual category, not an operational quantity**, and that is precisely why the last link remains open. An attempt to build that ordering — five distinct routes for distinguishing types of closure — is reported in the companion document (Companion Paper, v1.2), together with the exact point where it stops: none of the routes could be formalized as a coupling capable of entering the calculation of a frequency.

On circularity

A recurring objection to $T \equiv M-0$ is that it would be circular, or a disguised definition: no difference between time and motion has ever been found because time is measured *with* motion — a clock is counted motion — so the identity would be an artifact of our instruments.

The circularity cuts both ways: motion is not defined without time either, since a velocity is a change of position per unit time. Whoever holds that motion is primary falls into the same circularity they object to. But that only draws even: circularity is compatible with identity and also with the interdependence of two distinct, inseparable things.

What breaks the tie is the above. Interdependence does not predict that something without an internal cycle has zero proper time; identity together with the closure condition does.

The root tick Θ : how I write it without inventing “a clock”

Cosmological expansion can be expressed as an accumulated quantity. I take it as the root tick of Tempo.

$$\Theta = \ln(a/a_0) \quad (1)$$

$$\Theta = -\ln(1+z) \quad (2)$$

- a : scale factor
- a_0 : reference
- z : redshift

In this framework, it is not that Θ “measures” time: **Θ is the tick of Time.**

Cosmological scope: when I say “expansion” in T≡M, I mean the universe at large scales under the standard cosmological approximation of homogeneity/isotropy (FLRW-type). My “root tick” Θ is meant to describe the global average becoming of the universe, not local details.

Coordinates vs foundation: in cosmology one often writes $a(t)$, but in T≡M I do not take t as fundamental; I take it as a useful coordinate. The point of T≡M is interpretive: Θ represents the root tick of becoming, and usual time parametrizations are used as descriptive convenience, not as the primary ontological entity.

Nested motion

Let X be a representation of the physical state of the universe. I consider the elemental change dX without introducing “per unit time”.

To express the idea without committing to a strict mathematical sum, I use “+” as **conceptual composition** (not as literal numerical addition). If the reader prefers, it can be read as “combination” or “level-decomposition”.

$$dX = dX_{\text{root}} + dX_{\text{local}} \quad (3)$$

$$dX_{\text{local}} = \sum_{k \geq 1} dX_k \quad (4)$$

- dX_{root} is associated with the advance of Θ .
- dX_k represents motions at internal levels: from cosmological down to microscopic.

About the decomposition: the form (3)–(4) is not presented as a mathematical theorem nor as a strict sum of magnitudes. It is a conceptual map: root → levels → counts. My goal is to capture the hierarchy of motions as an interpretive structure of Tempo.

Minimal event $E_{\min}$

Functional definition

In this version I do not define $E_{\min}$ by saying “it is decoherence” or “it is some specific interaction.” That would commit to a microphysics I do not yet have.

Instead, I define it by its role:

Definition: $E_{\min}$ is the elemental event that increments a counter $N_{\min}$ and represents the minimal unit of physical activity required for locally observable change to exist within the root tick.

Minimal counter

I introduce $N_{\min}$ as an accumulated counter of minimal events.

Conjectures (what I want to be challenged on)

Strong conjecture — “Total pause”

Status of the strong conjecture: “total pause” is presented as a conceptual/ontological limit of $T \equiv M$. I do not offer it today as a direct laboratory prediction. The empirically discussable core is the weak conjecture below.

$$dN_{\min} = 0 \Rightarrow dX_k = 0 \text{ (for all } k) \quad (5)$$

Interpretation: if minimal activity is null, local motion is null at any level. There is no “empty time” left: there is **total pause**.

And the root conditioning:

$$d\Theta > 0 \Rightarrow dN_{\min} > 0 \quad (6)$$

Weak conjecture — Relation between clocks and the root tick

The testable path is to study the relation between local counts N_R and Θ .

Let R be a clock-process with counter N_R . Central object:

$$\frac{dN_R}{d\Theta} = F_R(\text{local conditions}) \quad (7)$$

What it would take to abandon $T \equiv M$

A framework that cannot lose says nothing. So I put in writing what result would make me abandon the premise.

The general criterion: a demonstration that time and motion are distinct. If one appeared, I would abandon the idea entirely.

Concrete forms it could take:

- **Proper time without internal motion.** A system in which proper time accumulates while nothing changes internally would refute $T \equiv M-0$ and $T \equiv M-4$ together. The symmetric case — motion without proper time — already exists and is the photon, but it counts in favor of the premise.
- **Time emerging from something other than motion.** Several quantum-gravity programs make time emerge from correlations between subsystems. If one of them were confirmed with a substrate not reducible to motion, the identity would not survive in its current form. *This does not contradict the affinity with Page and Wootters (1983) that I point out in the companion document: that time is not fundamental is compatible with $T \equiv M$; that it emerges from something not reducible to motion is not.*

- **Clocks that differ without a difference in motion.** A differential drift attributable to something other than internal dynamics or local conditions would leave the reading of clocks as nested sub-counts without support.

Two levels are easy to confuse and worth separating. The criteria above target the **premise**; bound (11) targets the **parameter**. The existence of the residual would not prove $T \equiv M$, and its absence would not refute it: ε_R is a minimal extension, not a necessary consequence. And absence of signal does not count as confirmation: if comparisons keep showing no drift, this framework remains what it is today, an ontological framework without a signature of its own.

Why the constant residual is unobservable (the v2.1 correction)

The v2.0 minimal ansatz read $dN_R/d\Theta = \kappa_R(1 + \varepsilon_R) + \delta_R$, with constant ε_R (the choice $G(\Theta) = 1$). Review showed that this form, while simplest, **cannot leave a trace in any real measurement**. The argument has two arms, and both close the door:

Arm 1 – universal residual. Metrology does not measure $dN_R/d\Theta$ directly; it measures **frequency ratios** ν_1/ν_2 between two clocks. Note that the lab-time ratio equals the Θ -ratio, because the conversion factor $d\Theta/dt$ cancels:

$$\frac{\nu_1}{\nu_2} = \frac{dN_1/dt}{dN_2/dt} = \frac{dN_1/d\Theta}{dN_2/d\Theta}$$

If ε_R is the same for all clocks (universal), the factor $(1 + \varepsilon_R)$ appears in numerator and denominator and **cancels exactly**. No ratio measurement, at any precision, can ever see it.

Arm 2 – clock-dependent residual. If ε_R differs per clock species but is constant, then $\kappa_R(1 + \varepsilon_R)$ is indistinguishable from a redefined intrinsic scale $\kappa'_R = \kappa_R(1 + \varepsilon_R)$. Since κ_R is not predicted from first principles but **calibrated**, the residual is silently absorbed into the calibration. Invisible again.

Conclusion: the $G(\Theta) = 1$ ansatz of v2.0 has no observable content. The honest move is not to defend it but to replace it. The minimal form that survives both arms is a residual that **changes with the advance of the root tick** – a drift.

Minimal ansatz, corrected (revised eq. 8)

$$\frac{dN_R}{d\Theta} = \kappa_R [1 + \varepsilon_R G(\Theta)] + \delta_R(\text{conditions}), \quad G(\Theta) = \Theta - \Theta_{\text{ref}} \quad (8)$$

- κ_R : intrinsic scale of clock R (calibrated).
- δ_R : corrections from local conditions (gravity, velocity, environment) – the known physics.
- ε_R : residual drift parameter (if it exists), per unit Θ .

Minimal choice, restated: in v2.0 I adopted $G(\Theta) = 1$ “to avoid arbitrariness.” v2.1 redefined what minimal means: the minimal ansatz is the simplest one **with observable content**, and that is a drift. A constant cannot be seen; a drift can.

Reference epoch: I take $\Theta_{\text{ref}} = 0$ at the present epoch ($a = a_0$), which $\Theta = \ln(a/a_0)$ already satisfies by construction. Writing $G(\Theta) = \Theta - \Theta_{\text{ref}}$ explicitly makes the ansatz manifestly invariant under shifts of the Θ origin: a shifted origin would otherwise reintroduce a constant term – exactly the absorbable case the correction removes. Only the drift is physical; the reference point is convention.

Clear reading:

- If $\varepsilon_R = 0$ for all clocks, T=M remains an ontological/synthesis framework.
- If a consistent drift $\varepsilon_R \neq 0$ appeared across configurations and epochs, T=M would gain physical content (and would demand formalization far beyond this manuscript).

The observable, and the first numerical bound

With the drift ansatz, take two clock species and form the measurable ratio.

Assumptions behind eq. (9), stated explicitly:

- **Small residuals:** the expression is leading-order in ε_R ; higher-order terms are negligible at current precision.
- **Known effects subtracted:** $\delta_R(\text{conditions})$ — gravity, velocity, environment, systematics — is modeled and removed, which is standard metrology practice.
- **Effective sensitivity absorbed:** different transitions respond with different sensitivity to any underlying drift; here those factors are absorbed into effective coefficients. A rigorous treatment propagates per-pair sensitivity coefficients, as Sherrill et al. (2023) do for constants variation.

Under these assumptions:

$$\frac{d}{d\Theta} \ln \frac{\nu_1}{\nu_2} \approx \varepsilon_{R,1} - \varepsilon_{R,2} \equiv \Delta\varepsilon_R \quad (9)$$

Only the **difference** between species survives — this is the honest limit of ratio metrology, stated below.

Converting to laboratory time with $d\Theta/dt = H(t) \approx H_0$ today:

$$\frac{d}{dt} \ln \frac{\nu_1}{\nu_2} \approx \Delta\varepsilon_R \cdot H_0 \quad (10)$$

Now the numbers. Long-term comparisons of optical frequency ratios constrain fractional drifts at the level of $\sim 10^{-18} \text{ yr}^{-1}$ (optical lattice clocks and clock networks; see Bloom et al. 2014, Beloy et al. / BACON Collaboration 2021, and the constants-variation analyses of Sherrill et al. 2023). Today's expansion rate is $H_0 \approx 70 \text{ km s}^{-1} \text{ Mpc}^{-1} \approx 7 \times 10^{-11} \text{ yr}^{-1}$ (the exact value, within the current ~ 67 – 73 range of the Hubble tension, does not affect an order-of-magnitude bound). Then:

$$|\Delta\varepsilon_R| \lesssim \frac{10^{-18} \text{ yr}^{-1}}{7 \times 10^{-11} \text{ yr}^{-1}} \approx 10^{-8} \quad (11)$$

This is the first numerical bound on the T=M residual: if the weak version of T=M exists as a species-dependent drift, it is smaller than about one part in a hundred million per unit of accumulated expansion. Using the strongest published constraints (drifts bounded near 10^{-19} yr^{-1}) tightens this proportionally, toward 10^{-9} .

Caveats, stated plainly:

- This is an **order-of-magnitude translation**, not a dedicated re-analysis. A rigorous bound requires propagating the sensitivity coefficients of each clock pair (different transitions respond differently to any underlying drift), exactly as Sherrill et al. (2023) do for variations of fundamental constants. That re-analysis is the natural next step and is explicitly invited.

- Operationally, a $\Delta\epsilon_R$ drift is **the same class of signal** as a temporal variation of fundamental constants. Existing bounds therefore transfer directly — and, symmetrically, a future detection could not be attributed to $T\equiv M$ rather than varying constants without additional structure. I state this overlap rather than hide it.
- For $T\equiv M$ to acquire a **signature of its own** someday, it would need a rule for how ϵ_R depends on the type of transition (electromagnetic vs. nuclear, different sensitivity classes). This manuscript does not have that rule and does not invent one. It is marked as the concrete direction in which the framework would stop being a reparametrization and start being physics. An attempt to derive it from the closure condition was carried out and did not arrive; the companion document reports where it stops and why.
- **The residual should not be read as wear.** The intuitive reading would be that expansion “stretches” local systems. It does not survive contact with known physics: a bound system does not undergo a continuous stretching proportional to H — its response is all-or-nothing after transients (Price and Romano, 2012) — the known local effect of a cosmological constant is a static offset and not a cumulative drift (Carrera and Giulini, 2010), and the result also holds under quantum treatment, where the regime is set by a characteristic timescale of the system (Lemos, 2014). ϵ_R should therefore be read as the parameter of a residual coupling of **unspecified origin** between the cosmological level and local dynamics.
- **The two-factor form is the standard one.** Eq. (9) decomposes the observable into a universal quantity times a coefficient that depends on the clock species. That is the usual structure in the phenomenology of locally coupled light scalar fields, where local tests do not bound the field’s velocity on its own but its product with per-species microscopic couplings (Trivedi, 2025). What matters for the program of this manuscript is that in that literature the per-species coefficient **has no independent experimental determination** and is estimated via sensitivity to fundamental constants. That is where an alternative rule, ordered by type of closure, would have content.

The honest limit: what ratios can never see

Equation (9) makes something explicit that v2.0 did not say: **frequency ratios are blind to the universal part of ϵ_R** . If every clock in the universe drifted identically with Θ , every ratio would stay perfectly constant and no laboratory comparison would ever notice.

Constraining a universal residual would require comparing local counts against a **non-clock reference** — cosmological observables tied to Θ itself (redshift drift programs, for example). That is far beyond this manuscript, and I flag it as the hard, long-term branch of the weak program.

A further caution is required here, and I state it rather than hide it: a strictly universal drift could be physically equivalent to a **pure reparametrization**. If every process drifts identically, no dimensionless relation between observables changes — and only dimensionless relations are unambiguously physical. The universal branch may therefore be not merely unobservable in ratios, but potentially empty of physical content. Whether it is a real degree of freedom or a relabeling of units is itself an open question of the program.

So the weak version of $T\equiv M$ splits cleanly:

- **Species-dependent drift:** observable in ratios; bounded today at $|\Delta\epsilon_R| \lesssim 10^{-8}$ (eq. 11).
- **Universal drift:** unobservable in ratios by construction; open in principle, untested today.

A theory that says out loud where it cannot be seen is worth more than one that hides it.

Figure 1 (conceptual toy model)

Toy model (conceptual):

- **Clock A:** $N_A(\Theta) = \kappa_A (\Theta + \frac{1}{2}\varepsilon_A\Theta^2)$
- **Clock B:** $N_B(\Theta) = \kappa_B (\Theta + \frac{1}{2}\varepsilon_B\Theta^2) + \epsilon g(\text{environment})$
- **Clock C:** $N_C(\Theta) = \kappa_C \Theta$ (reference clock; its residual is set to zero *by convention*, since only differences between clocks are observable — see eq. 9)

More concrete version (still schematic):

- R can be an atomic clock (Cs, Sr, Yb), a pulsar, or a stable nuclear process.
- The relevant comparison is not “seconds,” but how the **ratios of counts** evolve with respect to Θ .

How it could be tested (minimal program, updated)

If someone with expertise wanted to test the weak version of $T \equiv M$, the program is now sharper than in v2.0:

1. **Choose concrete clock pairs R_1, R_2 :** different species (e.g., Sr vs Yb, Cs vs optical), because only differential residuals are visible.
2. **Also include pairs within a single species.** Two different transitions of the same ion share nucleus, environment, gravitational potential, and motion, so that corrections from local conditions cancel in common mode. It is the most controlled case available, and long-term data exist (Godun et al., 2014; Lange et al., 2021).
3. **Practical observable:** the frequency ratio ν_1/ν_2 and its **long-term drift** (not just its stability).
4. **Base model:** compute/fit all known effects (gravity, velocity, environment, systematics). This corresponds to $\delta_R(\text{conditions})$.
5. **Minimal $T \equiv M$ fit:** fit a residual linear drift in Θ — equivalently, a drift rate $\Delta\varepsilon_R \cdot H_0$ in laboratory time (eq. 10).
6. **Outcome A:** $\Delta\varepsilon_R$ is bounded ever more tightly ($T \equiv M$ as ontology + quantitative empirical bounds — the current state, eq. 11).
7. **Outcome B:** a consistent residual drift appears across pairs and epochs, with a pattern not attributable to known constants-variation channels (which would be extraordinary and would demand real formalization).

Note: “no signal” + tighter bounds is also a result. It is, in fact, the result this manuscript already delivers at order-of-magnitude level.

Compatibility with standard relativity/cosmology (interpretation)

I do not intend to replace standard cosmology or general relativity. I propose an ontological reading:

- Θ as the root tick.
- Clocks as nested sub-counts.
- A weak program (8) as a possible residual extension — in drift form, and bounded.

Methodological transparency and call for review

The correction at the heart of v2.1 — the unobservability of the constant residual, and the translation of existing bounds into a first number — came out of a careful review of v2.0. I adopted it because the argument stands on its own, not because of where it came from. The same holds for this v2.2: the closure condition and the acknowledgment of its limits came out of a review process that included trying to derive a signature of its own and documenting where that attempt stopped.

I present this as an open preprint and welcome corrections, counterexamples, and review, especially regarding:

- The cancellation argument (arm 1 / arm 2): is there a ratio-accessible effect of a constant residual that I missed?
 - The translation (9)–(11): proper treatment of sensitivity coefficients per clock pair.
 - How to design a non-ratio test for the universal branch.
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Closing (my way of saying it)

In $T \equiv M$, the universe “Tempo-ed” from its beginning: by expanding, it initiated the root tick. Local motions do not occur “in” time; they constitute time as motions nested within cosmic becoming.

v2.0 planted a parameter. v2.1 learned where that parameter can and cannot hide, and put the first number on it. If the residual exists as a difference between clocks, it is smaller than one part in a hundred million per unit of expansion. If it is universal, it hides where ratios cannot reach — and now that hiding place is named.

v2.2 wrote down why that parameter is a natural extension of the closure condition, and made explicit the physics that would have to determine it.

The seed is planted. The Tempo will tell.

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Appendix — Minimal notation

Symbol	Meaning
$\Theta = \ln(a/a_0) = -\ln(1+z)$	Root tick (cosmological Tempo)
a	Cosmological scale factor
z	Redshift
X, dX	Universe state; elemental change
$E_{\min}, N_{\min}$	Minimal event; minimal counter
R, N_R	Local clock-process; process count
ε_R	Residual drift parameter, per unit Θ (if it exists); denoted α_R in v2.0
$\Delta\varepsilon_R$	Differential residual between two clock species
$G(\Theta)$	Residual structure function; v2.1 adopts $G(\Theta) = \Theta - \Theta_{\text{ref}}$, with $\Theta_{\text{ref}} = 0$ today
κ_R	Intrinsic scale of clock R (calibrated)
δ_R	Corrections from local conditions
$H_0 \approx 7 \times 10^{-11} \text{ yr}^{-1}$	Present expansion rate (conversion $d\Theta/dt$)
Closure	Condition in which the internal relations of a motion sustain one another, forming a unit (T≡M-4). Required for a motion to constitute proper time
τ	Proper time: the count accumulated by a motion that closes. Zero along null trajectories

Numbered equations

- (1)–(2) definition of Θ (root tick)
- (3)–(4) nested-motion structure
- (5)–(6) strong conjecture (total pause and root conditioning)
- (7) general clock–root relation
- (8) minimal ansatz with observable content (drift form, $G(\Theta) = \Theta - \Theta_{\text{ref}}$)
- (9) ratio observable: differential drift $\Delta\varepsilon_R$
- (10) laboratory-time translation via H_0

- (11) first numerical bound: $|\Delta\varepsilon_R| \lesssim 10^{-8}$

Suggested citation: Moreira, M. (2026). *T≡M Theory — Time Is Motion: Cosmological Expansion as the Root Tick* (v2.2) [Preprint]. Zenodo.